PRACTICAL NO. 8

A system has RAM which can store four pages simultaneously to satisfy page requests. Write a Program to calculate percentage page hit and page fault if the system uses Least recently used page replacement technique for the page references entered by the user i.e. 1, 2, 3, 4, 1, 2, 5, 1, 2, 3, 4, 5 (No of Frames=3).

CODE:

```c
#include <stdio.h>

int main() {
    int frames[3], time[3];
    int pages[] = {1,2,3,4,1,2,5,1,2,3,4,5};
    int n = 12, i, j, pos;
    int faults = 0, hits = 0, counter = 0;
    for(i = 0; i < 3; i++) {
        frames[i] = -1;
        time[i] = 0;
    }
    for(i = 0; i < n; i++) {
        int found = 0;
        for(j = 0; j < 3; j++) {
            if(frames[j] == pages[i]) {
                hits++;
                counter++;
                time[j] = counter;
                found = 1;
                break;
            }
        }
        if(!found) {
            int min = time[0];
            pos = 0;
            for(j = 1; j < 3; j++) {
```

```c
                time[j] = counter;
                found = 1;
                break;
            }
        }
        if(!found) {
            int min = time[0];
            pos = 0;
            for(j = 1; j < 3; j++) {
                if(time[j] < min) {
                    min = time[j];
                    pos = j;
                }
            }
            frames[pos] = pages[i];
            counter++;
            time[pos] = counter;
            faults++;
        }
    }
    printf("Page Faults = %d\n", faults);
    printf("Page Hits = %d\n", hits);
    printf("Page Fault Percentage = %.2f%%\n", (faults*100.0)/n);
    printf("Page Hit Percentage = %.2f%%\n", (hits*100.0)/n);
    return 0;
}
```

OUTPUT:

```
Page Faults = 10
Page Hits = 2
Page Fault Percentage = 83.33%
Page Hit Percentage = 16.67%


=== Code Execution Successful ===
```